

ECON 3201

Econometrics for Economics and Finance

Assignment 1
Due Thursday October 8, 2026

Maxim

Instructions

Answer all questions. Please complete your assignment using Quarto. Submit both your Quarto file and pdf output file.

Theory Questions

Q1

Create a personal Quarto website hosted on GitHub pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

- i. An “Index” or “Home” page. This page should include the following:
 - Your name
 - Your photo (optional)
 - Your credentials
 - An “About Me” section that provides a brief (one paragraph) introduction to yourself. [10 Marks]
- ii. A blog page. This will be the landing site for future assignments [5 Marks]
- iii. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
- iv. Navigation bar with “Home” and “Blog” buttons. The navigation bar is edited in the “_quarto.yml” file. [2 Marks]

[Link to website](#)

Q2

Write your solutions using LaTeX. Show your work. You may use R to compute the answers.

Let $X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a uniformly distributed variable with probability distribution function $f_X(x) = \frac{1}{12}$.

(a) Check that $f_X(x)$ is a true probability distribution.

Each outcome in the sample space has probability $\frac{1}{12}$ of occurring. Since there are 12 outcomes in the sample space, it follows that the sum of probabilities adds up to 1. Thus, $f_X(x)$ is a true probability distribution.

(b) Compute the expected value of X .

$$1 \times \frac{1}{12} + 2 \times \frac{1}{12} + \dots + 11 \times \frac{1}{12} + 12 \times \frac{1}{12} = 6.5$$

The expected value of X is 6.5.

(c) Compute the variance of X .

$$(1 - 6.5)^2 \times \frac{1}{12} + (2 - 6.5)^2 \times \frac{1}{12} + \dots + (11 - 6.5)^2 \times \frac{1}{12} + (12 - 6.5)^2 \times \frac{1}{12} = 13$$

The variance of X is 13.

Q3

Let $X \sim N(4, 5)$. Find the probability of each event:

(a) $Pr(X \leq 6)$

$$Pr(X \leq 6) = 0.8144533$$

(b) $Pr(X > 4)$

$$Pr(X > 4) = 1 - Pr(X \leq 4) = 0.5$$

(c) $Pr(-1 \leq X < 1)$

$$Pr(-1 \leq X < 1) = Pr(X < 1) - Pr(-1 \leq X) = 0.07718259$$

Q4

Let $Z \sim N(0, 1)$.

(a) Compute $Pr(Z \leq 0.83)$

$$Pr(Z \leq 0.83) = 0.7967306$$

(b) Compute $Pr(-0.83 \leq Z \leq 0.83)$

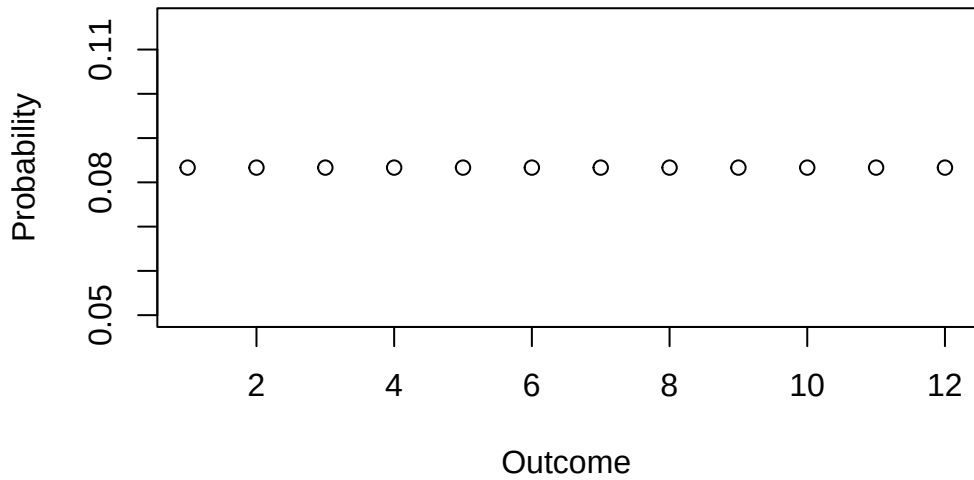
$$Pr(-0.83 \leq Z \leq 0.83) = Pr(Z \leq 0.83) - Pr(-0.83 \leq Z) = 0.5934612$$

R Exercises**Q5**

Let $X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

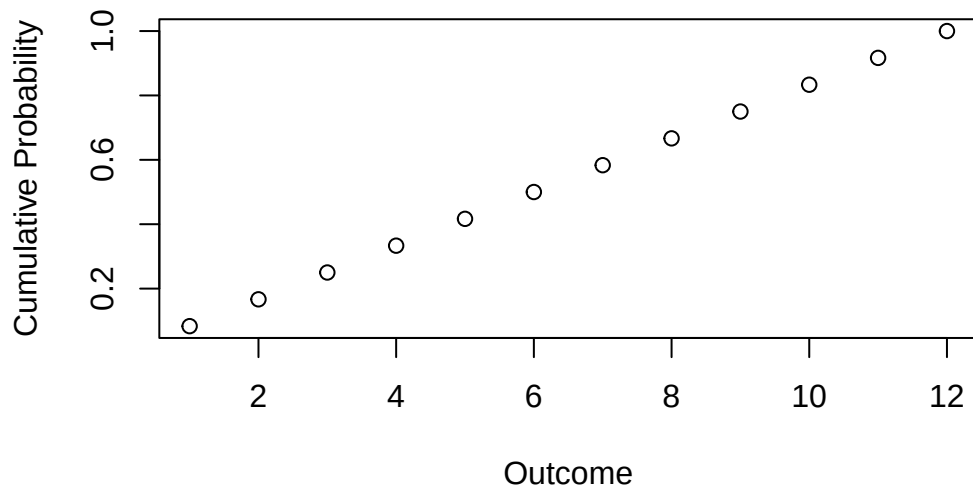
(a) Plot the probability distribution function $f_X(x)$ in R.

```
p <- rep(1/12, 12)
plot(p, xlab = "Outcome", ylab = "Probability")
```



(b) Plot the cumulative probability distribution function $f_X(x)$ in R .

```
p <- rep(1/12, 12)
c <- cumsum(p)
plot(c, xlab = "Outcome", ylab = "Cumulative Probability")
```



(c) Compute the expected value of X in R.

```
x <- 1:12  
mean(x)
```

```
[1] 6.5
```

$E(X) = 6.5$

(d) Compute the variance of X in R.

```
x <- 1:12  
var(x)
```

```
[1] 13
```

$Var(X) = 13$

Q6

The standard normal distribution function is given by

$$f(x) = \frac{e^{-(x^2/2)}}{\sqrt{2\pi\sigma}}$$

- (a) Define this function in R. (Note: π is coded as `pi` and e is coded as `exp`.)

```
sndf <- function(x) {  
  result <- exp(1)^(-(x^2/2))/sqrt(2 * pi)  
  return(result)  
}
```

- (b) Integrate this function over the limit $[-\infty, \infty]$ using the `integrate()` command in R. Infinity in R is written as `Inf`. To evaluate the integral, use `integrate()$value`

```
integrate(sndf, -Inf, Inf)$value
```

```
[1] 1
```

- (c) Compute the expected value of the standard normal distribution in R and confirm it is equal to 0.

```
integrate(function(x) x * sndf(x), -Inf, Inf)$value
```

```
[1] 0
```

The expected value of this function is equal to 0.

- (d) Compute the variance of the standard normal distribution in R and confirm it is equal to 1.

```
integrate(function(x) x^2 * sndf(x), -Inf, Inf)$value
```

```
[1] 1
```

The variance of this function is equal to 1.

- (e) Compute $Pr(Z \leq 0.83)$

```
integrate(sndf, -Inf, 0.83)$value
```

```
[1] 0.7967306
```

$Pr(Z \leq 0.83) = 0.7967306$

(f) Compute $Pr(-0.83 \leq Z \leq 0.83)$

```
integrate(sndf, -Inf, 0.83)$value - integrate(sndf, -Inf, -0.83)$value
```

```
[1] 0.5934612
```

$Pr(-0.83 \leq Z \leq 0.83) = 0.5934612$

(g) Compute $Pr(-0.83 \leq Z \leq 0.83)$ using the `pnorm()` command and compare your results to (f).

```
pnorm(0.83) - pnorm(-0.83)
```

```
[1] 0.5934612
```

$Pr(-0.83 \leq Z \leq 0.83) = 0.5934612$

The result of using `pnorm()` is the same as the result in (f).

Q7

Let X denote the prison sentence, in years, for people convicted of auto theft in a particular province in Canada. Suppose the PDF of X is given by:

$$f(X) = \frac{1}{9}x^2, \quad 0 < x < 3.$$

(a) Define $f(x)$ in R.

```
f <- function(x) {  
  result <- (1/9) * x^2  
  return(result)  
}
```

(b) Integrate $f(x)$ over the interval $[0,3]$.

```
integrate(f, 0, 3)$value
```

```
[1] 1
```

(c) Compute the expected prison sentence.

```
integrate(function(x) x * f(x), 0, 3)$value
```

```
[1] 2.25
```

The expected prison sentence is 2.25 years.

Q8

Let X follow a Student t distribution with M degrees of freedom. Then

$$X = \frac{Z}{\sqrt{W/M}} \sim t_M,$$

where Z is a standard normal variable and W is a chi-squared variable with M degrees of freedom.

(a) On the same figure plot the following:

- (i) Plot the standard normal probability distribution using `curve()`. Specify the range as $[-5,5]$ using `xlim`. Name the `ylab` "Density". The main title should be "Densities of t Distribution".
- (ii) Plot the Student t distribution with 5 degrees of freedom using `curve()` and `dt()`. Specify the range as $[-5,5]$ using `xlim`. Choose a different colour, i.e. `col=2`. To learn more about `dt`, type `?dt` in your console.
- (iii) Plot the Student t distribution with 15 degrees of freedom using `curve()` and `dt()`. Specify the range as $[-5,5]$ using `xlim`. Choose a different colour, i.e. `col=3`. To learn more about `dt`, type `?dt` in your console.
- (iv) Plot the Student t distribution with 25 degrees of freedom using `curve()` and `dt()`. Specify the range as $[-5,5]$ using `xlim`. Choose a different colour, i.e. `col=4`. To learn more about `dt`, type `?dt` in your console.
- (v) Include the following legend in your plot:

```
legend("topright", c("N(0, 1)", "M=2", "M=4", "M=25"), col = 1:4, lty = c(1, 1, 1, 1))
```



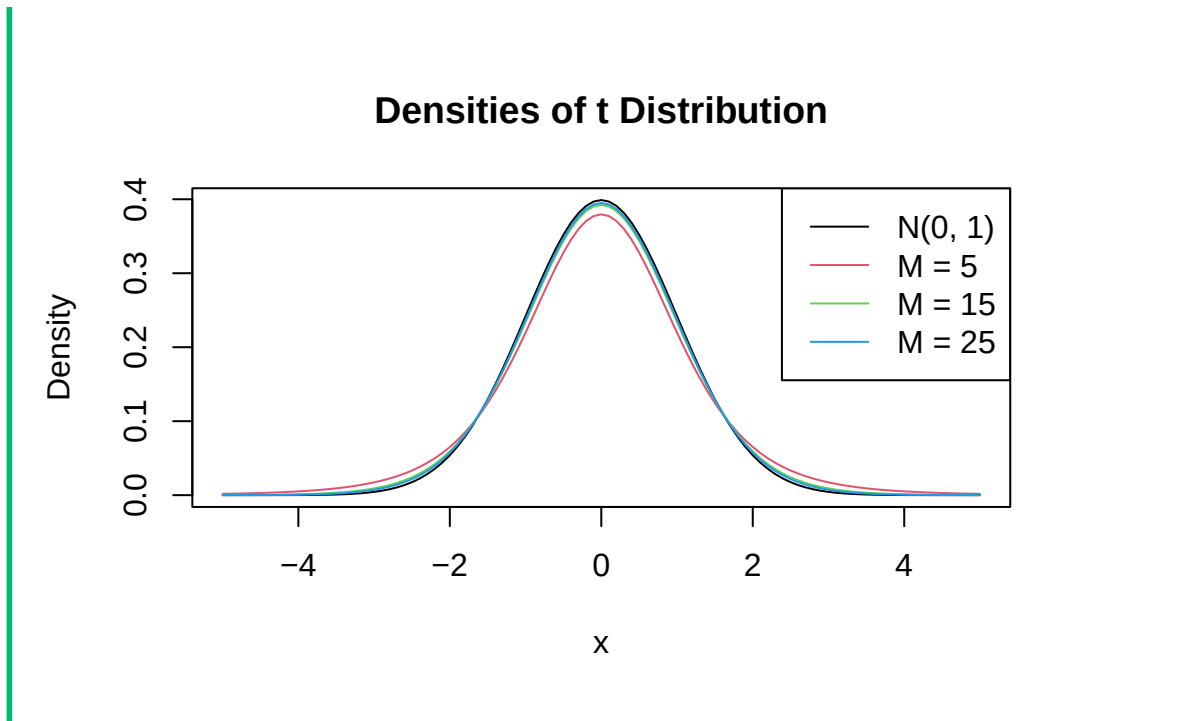
```
curve(dnorm(x),
      main = "Densities of t Distribution",
      ylab = "Density",
      xlim = c(-5, 5))

curve(dt(x, 5, log = FALSE),
      add = TRUE,
      xlim = c(-5, 5),
      col = 2)

curve(dt(x, 15, log = FALSE),
      add = TRUE,
      xlim = c(-5, 5),
      col = 3)

curve(dt(x, 25, log = FALSE),
      add = TRUE,
      xlim = c(-5, 5),
      col = 4)

legend("topright",
      c("N(0, 1)",
        "M = 5",
        "M = 15",
        "M = 25"),
      col = 1:4,
      lty = c(1, 1, 1, 1))
```



(b) What do you observe as the degrees of freedom increase?

As the degrees of freedom increase, the student t distribution approaches that of the standard normal distribution.